

8.1 DISTINCTION TASK (Machine Learning Mini Project)

About this task

Step-1

This task is designed to assess the Distinction level expectations. You will submit a concise PDF report and a separate Jupyter notebook code file (.ipynb) for completing this credit task.

Step-2

Your tutor will then review your submission and will give you feedback. If your submission is incomplete the tutor will ask you to include missing parts. As part of assurance of learning (AoL) in the GenAI era, tutors may ask follow-up questions (via OnTrack or Viva Voce) to clarify elements of your submission and assess your understanding of relevant topics.

Feedback and submission deadlines

Feedback deadline: 18/09/2026 (No submission before this date means no feedback!)

Submission deadline: Before creating and submitting portfolio.

Evidence of Learning: ML Mini Project

Mini Project: Sydney Housing Price Prediction and Decision Support System

Project Overview

Property valuation is one of the most important applications of machine learning in real estate analytics. In this project, you will act as a data scientist engaged by a real estate agency to develop a housing price prediction and decision support system. The project requires you to complete the full machine learning lifecycle, from collecting and preparing data to building predictive models, interpreting their results, deploying a prototype application, and evaluating the role of machine learning in property valuation.

The primary objective of this project is not simply to achieve the highest predictive accuracy. Rather, you are expected to demonstrate your ability to make informed decisions throughout

the machine learning process, justify those decisions using evidence, critically evaluate model performance, and communicate your findings to both technical and non-technical audiences.

Part 1: Problem Definition and Data Collection

Begin by selecting three Sydney suburbs in which you would personally consider purchasing a property. The selected suburbs should represent substantially different housing markets. For example, they may differ in location, demographic characteristics, accessibility, or property values. Using publicly available information from [realestate.com.au](https://www.realestate.com.au), [domain.com.au](https://www.domain.com.au), or a similar property website, manually construct a dataset containing **at least** 100 sold properties, with a minimum of 30 properties from each suburb.

For each property, collect the sale price (the prediction target) along with a broad range of property characteristics/features that may influence housing value. Advanced ML users/students are encouraged to incorporate additional data information such as real estate agent descriptions (i.e., text data) of the property, to enhance their analysis and predictive modelling. You should then introduce the housing price prediction problem, explain the motivation for your suburb selection, and discuss how the features of the chosen suburbs may influence housing prices. You should also discuss the quality of the collected dataset, challenges encountered during data collection, missing information, potential sources of bias, and limitations that may affect the validity of your analysis.

Part 2: Data Understanding and Feature Engineering

Before building predictive models, develop a comprehensive understanding of the dataset. Explore the distribution of housing prices, investigate differences between suburbs, examine trends over time, and identify unusual observations or potential outliers. Visualisations should be used to support your analysis and communicate important findings.

Next, consider how additional features or data may improve predictive performance. Before creating any engineered features, identify the three variables that you believe will have the strongest influence on housing prices and justify your reasoning. Reflect on whether your feature engineering decisions align with your initial understanding of the housing market.

Part 3: Model Development and Evaluation

Develop three regression models that represent different modelling approaches. Before training any model, explain why each model has been selected and predict which model you expect to perform best. Your discussion should consider the characteristics of the dataset as well as the strengths and weaknesses of each modelling approach.

Train and evaluate the models using k-fold cross-validation and appropriate regression metrics. The evaluation should go beyond reporting numerical results and should include a critical analysis of model behaviour. Consider whether any models appear to be underfitting or overfitting and discuss how model complexity influences performance. Once the evaluation is complete, revisit your original expectations and determine whether they were supported by the experimental results. Conclude this part by recommending the most appropriate model for the problem and providing a clear justification for your decision.

Part 4: Investigating Prediction Failures

Machine learning models often reveal their limitations through their largest errors. Identify the five properties that produced the largest prediction errors and investigate each case in

detail. Consider the characteristics of the property, the magnitude of the prediction error, and any contextual factors that may explain why the model failed.

Using these examples, discuss the limitations of your modelling approach and identify information that was unavailable but may have improved predictive performance. Reflect on situations in which housing price predictions should be interpreted cautiously and evaluate whether some types of properties are inherently more difficult to model than others.

Part 5: Final Deployment and Reflection

Develop a simple web-based application that allows users to enter or upload property features and information and receive a predicted sale price from your developed and deployed ML model. The application may be implemented using Streamlit, Gradio, Flask, or another suitable framework. In your report, describe how the application was developed and attach instructions for using the application with **clear screenshots**.

Conclude the project with a reflection on the complete machine learning workflow. Discuss the key lessons learned, the challenges encountered during data collection, preprocessing, feature engineering, model development, and evaluation, as well as the trade-offs between predictive performance and practical deployment. Also reflect on the potential biases, fairness concerns, and limitations of the data and models used, as well as how the project could be improved if additional data, resources, computational power, or time were available.

Suggested report length: 1,000 words (excluding code, tables, figures, and **references**).

Deliverables: Submit a **PDF report** that addresses all requirements in Parts 1–5. The report should include all relevant code, outputs, tables, figures, analysis, screenshots, references, a description of how the application was developed, and instructions on how to build, run, and use the code and application. The report should also include an acknowledgement of any use of GenAI (if applicable). In addition, in the report **include an accessible Microsoft OneDrive or Dropbox link** to a ZIP archive containing the collected housing dataset, all source code and notebooks including the deployed application source code. Alternatively, you can **include an accessible GitHub link** to a created GitHub project containing your data and code with brief descriptions. All deliverables must be submitted via **OnTrack** (<https://ontrack.deakin.edu.au>) and should be sufficiently documented to enable the reproduction of results and deployment of the application. For further details, refer to the submission requirements for the Pass and Credit tasks.

Note: Ensure that all code runs successfully from start to finish without errors and that all outputs are visible and reproducible at the time of submission.

Note on Using GenAI in this assessment

You are not required or expected to use GenAI for this task. However, to support your learning about the ethical and responsible use of GenAI including acting with academic integrity, **open and acknowledged** use of GenAI is acceptable in this assessment task. You may use GenAI in this task specifically for *planning, brainstorming, and editing your work for grammatical and spelling errors*. **You must not directly copy and submit AI-generated content**. Any use of AI-generated content must be critically reviewed, understood, and appropriately modified. Submissions will be reviewed by tutors and relevant assessment tools.

Your final submission must represent your own work and clearly demonstrate your understanding, critical thinking, and application of the concepts learned in this unit.

If you use GenAI for this assessment task, you should:

- [Evaluate the accuracy and quality](#) of any GenAI generated material before you try to adapt it. You remain fully responsible for the accuracy, integrity, and quality of your final submission.
- [Acknowledge how you used GenAI](#) tools **in your submission** (including report and codes), to show how you are demonstrating learning you have gained in the unit, making informed decisions about this learning, and **acting with integrity**.